

The planes over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are 4-chromatic

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ABSTRACT. For a subfield $F \subset \mathbb{R}$ let $\chi(F^2)$ be the chromatic number of the graph on F^2 joining points at Euclidean distance 1. We show that $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$, answering a question left open by Moorhouse (2010), Madore (2015) and Exoo–Ismailescu (2018), and that $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$. Voronov conjectured both values in Polymath16 (2021). The upper bounds come from Madore’s reduction argument, run in other coordinates. With $\alpha = x + y/\sqrt{3}$ and $\beta = 2y/\sqrt{3}$ the squared distance becomes $\alpha^2 - \alpha\beta + \beta^2$, a form that is anisotropic modulo any prime above 2 with residue field $\mathbb{F}_2$. Speyer used reduction modulo 2 in these coordinates in 2018 to 4-colour the Moser ring. The observation here is that, over these two fields, it colours the whole plane.

1. Introduction

The Hadwiger–Nelson problem asks for $\chi(\mathbb{R}^2)$. Since de Grey [1] and Exoo–Ismailescu [3] it is known that $\chi(\mathbb{R}^2) \in \{5, 6, 7\}$; see Soifer [10] for the history. For a subfield $F \subset \mathbb{R}$ one has $\chi(F^2) \leq \chi(\mathbb{R}^2)$. Known values include:

- $\chi(\mathbb{Q}^2) = 2$ (Woodall [12]);
- $\chi(\mathbb{Q}(\sqrt{2})^2) = 2$, $\chi(\mathbb{Q}(\sqrt{3})^2) = 3$ and $\chi(\mathbb{Q}(\sqrt{7})^2) = 3$ (Madore [5]);
- $4 \leq \chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) \leq 5$ (Madore [5], Prop. 4.6).

The field $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ is the smallest whose plane contains a Moser spindle. Moorhouse [6] noted that its exact value was undetermined, and Exoo and Ismailescu [3] asked whether a 5-chromatic unit-distance graph embeds in $\mathbb{Q}[\sqrt{3}, \sqrt{11}]^2$. In Polymath16, Voronov wrote that $\chi(\mathbb{Q}(i, \sqrt{3}, \sqrt{11})) = 4$ “seems likely, and the same is true in case (2, 3)”, but that nobody had proved it [9]. Case (2, 3) is the field $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}) = \mathbb{Q}(\zeta_{24})$. For it he suggested a colouring through a homomorphism $\mathbb{Z}[\zeta_{24}, 1/3] \rightarrow (\mathbb{Z}/4)[\zeta_{24}]$, adding “Perhaps there is a simpler way” [9].

Theorem 1. $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = \chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$. In particular, no 5-chromatic unit-distance graph has all its coordinates in $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, or all in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

The upper bounds are instances of the following criterion.

Theorem 2. Let $L \subset \mathbb{R}$ be a number field containing $\sqrt{3}$, and suppose some prime P of L above 2 has residue field $O_L/P \cong \mathbb{F}_2$. Then $\chi(L^2) \leq 4$.

Theorem 2 is Madore’s Proposition 3.2 [5] with the form $\alpha^2 - \alpha\beta + \beta^2$ in place of $x^2 + y^2$; his proof applies verbatim (compare Moorhouse [6], Lemma 4.2 and Theorem 7.1). In the coordinates (x, y) the argument fails at 2: the form $x^2 + y^2 \equiv (x + y)^2$ is isotropic modulo 2, and unit vectors such as $(1/2, \sqrt{3}/2)$ are not even integral at 2. Since $\sqrt{3} \in L$, we have $L(i) = L(\omega)$ with $\omega = e^{2\pi i/3}$, and the hypothesis on P implies that P is inert in $L(i)$.

2. The identity and the lemma

Put $Q(\alpha, \beta) = \alpha^2 - \alpha\beta + \beta^2$. For $x, y \in L$ set

$$\alpha = x + y/\sqrt{3}, \quad \beta = 2y/\sqrt{3}, \quad \text{so that} \quad x^2 + y^2 = Q(\alpha, \beta).$$

The identity is a direct expansion. Equivalently, $x + iy = \alpha + \beta\omega$, and Q is the norm form of $\mathbb{Q}(\omega)$. The map $(x, y) \mapsto (\alpha, \beta)$ is L -linear and bijective.

Lemma. Let v be the normalised valuation of L at a prime P above 2 with residue field $\mathbb{F}_2$. Then for all $(\alpha, \beta) \neq (0, 0)$ in L^2 , $v(Q(\alpha, \beta)) = 2 \min(v(\alpha), v(\beta))$.

Proof. Let $k = \min(v(\alpha), v(\beta))$ and let π be a uniformiser. Write $\alpha = \pi^k \alpha'$, $\beta = \pi^k \beta'$, with α', β' integral at P and not both in P . Then $Q(\alpha, \beta) = \pi^{2k} Q(\alpha', \beta')$. Modulo P , $Q(\alpha', \beta')$ reduces to $s^2 + st + t^2$ with $(s, t) \in \mathbb{F}_2^2 \setminus \{0\}$, since signs do not matter in characteristic 2. This value is 1 for every such (s, t) , because $u^2 + u + 1$ has no root in $\mathbb{F}_2$. Hence $v(Q(\alpha', \beta')) = 0$. ■

3. Proof of Theorem 2

Let $O_P = \{c \in L : v(c) \geq 0\}$ and let $\rho : O_P \rightarrow \mathbb{F}_2$ be reduction modulo P . Fix a set of representatives of the cosets L/O_P , and write $r(c)$ for the representative of $c + O_P$. Since L is countable, no choice principle is needed. Colour $(x, y) \in L^2$ by

$$c(x, y) = (\rho(\alpha - r(\alpha)), \rho(\beta - r(\beta))) \in \mathbb{F}_2^2,$$

which uses four colours. Suppose (x, y) and (x', y') are at distance 1, and let $\Delta\alpha, \Delta\beta$ be the differences of their α - and β -coordinates.

1. By linearity, $Q(\Delta\alpha, \Delta\beta) = 1$.
2. By the Lemma, $\min(v(\Delta\alpha), v(\Delta\beta)) = 0$. So $\Delta\alpha$ and $\Delta\beta$ lie in O_P and are not both in P .
3. Hence $r(\alpha) = r(\alpha')$ and $r(\beta) = r(\beta')$.
4. So the two colours differ by $(\rho(\Delta\alpha), \rho(\Delta\beta)) \neq (0, 0)$. ■

4. Proof of Theorem 1

Upper bounds. Both fields satisfy the hypotheses of Theorem 2.

- **$\mathbb{Q}(\sqrt{3}, \sqrt{11})$.** The prime 2 ramifies in $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{11})$ ($3, 11 \equiv 3 \pmod{4}$) and splits in $\mathbb{Q}(\sqrt{33})$ ($33 \equiv 1 \pmod{8}$). So $2 = P_1^2 P_2^2$, and both primes have residue field $\mathbb{F}_2$.
- **$\mathbb{Q}(\sqrt{2}, \sqrt{3})$.** The prime 2 ramifies in all three quadratic subfields, so $2 = P^4$ with residue field $\mathbb{F}_2$.

Both decompositions were also confirmed with PARI/GP and with sympy.

Lower bounds. For $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ the Moser spindle suffices. Take the unit rhombus with vertices $O, (1, 0), (1/2, \sqrt{3}/2), (3/2, \sqrt{3}/2)$, together with its image under the rotation with $\cos \theta = 5/6$, $\sin \theta = \sqrt{11}/6$. The two far tips are then at distance 1.

For $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, a 4-chromatic graph is already implicit in Voronov, Neopryatnaya and Dergachev [11]: their second series of 5-chromatic graphs is built from the 4-chromatic 10-vertex graph $L_{10,2}$ and the unit vectors ζ_{24} and $(\sqrt{6} + i\sqrt{3})/3$, all with coordinates in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. For completeness, here is a short explicit example. Take the unit vectors

$$u_1 = (1, 0), \quad u_2 = (0, 1), \quad u_3 = (-2/3 - \sqrt{2}/6, -2/3 + \sqrt{2}/6),$$

for which $|u_1 + u_2 + u_3|^2 = 1/3$.

Let $P_0 = O$ and $P_k = \sqrt{3}(u_1 + \dots + u_k)$. Join P_{k-1} and P_k by the unit rhombus whose middle vertices are $(P_{k-1} + P_k)/2 \pm u_k^\perp/2$. The result has 10 vertices and 16 edges.

1. In any 3-colouring, the two middle vertices of each rhombus take two distinct colours, and its two tips both take the third. So each rhombus forces its tips alike, and P_0 and P_3 share a colour.
2. But $|P_0P_3| = 1$, a contradiction. ■

5. Remarks

1. **Prior work.** Reduction modulo 2 in the coordinates $\alpha + \beta\omega$ goes back to Speyer [7]. In April 2018 he 4-coloured the Moser ring $R = O_K[1/3]$, $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, through $R/2R \cong \mathbb{F}_4 \times \mathbb{F}_4$. A computer search by Hubai, reported by Gibbs, then found that all its 4-colourings have period 8 [8], and Dúcz [2] has recently given geometric 4-colourings of the Moser lattice and ring. The ingredients of Theorem 2 are standard. The passage from a ring to the whole field by cosets is Madore's and Moorhouse's, and the integrality of unit vectors is the compactness of the norm-one torus at a place that does not split. What seems to have been missed is that the form $\alpha^2 - \alpha\beta + \beta^2$ makes the argument work at 2. With it, Speyer's colouring extends to every point and every unit vector of the plane over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, and the same construction works over $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.
2. **Consistency check.** The criterion must fail for fields that contain 5-chromatic graphs, and it does. In $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{247})$, the field of Exoo-Ismailescu, and in $\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{11})$, the field of Heule [4], every prime above 2 has residue field containing $\mathbb{F}_4$, where $u^2 + u + 1$ has roots.

6. Verification and provenance

- **Upper bounds.** Besides the proof, the colourings were tested by computer on random unit vectors $t/\bar{t}$ of each field and on unit-distance graphs with up to about 12 000 vertices, with no monochromatic edge. A second check, with separate code, its own 2-adic model and PARI/GP for the prime decompositions, gave the same results.
- **Lower bounds.** Both graphs can be checked by hand. An exhaustive computer search also confirms that neither has a proper 3-colouring.
- **Code and data.** Available from the author on request.

* These results were obtained with the assistance of an AI system (Claude, by Anthropic), which carried out the computations and the checks and helped to draft this note. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

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